

Abstract

- Smart glasses that enable live, two-way, frictionless communication across languages.
- The glasses listen for nearby speech, recognize the language, translate it into the user’s native language, and present the translation on a display visible only to the user.
- Additionally, the AI component utilizes relevant context and user data to generate a suggested response to be shown on the display, phonetically typed and accompanied by a translation into the user’s native language.
- The product includes a heads-up display (HUD), an integrated microphone, a long-lasting battery, and a high-performance Bluetooth processor, as well as a mobile companion app that allows users to manage their accounts, access device settings, and view previous conversations for learning purposes (Figure 1).

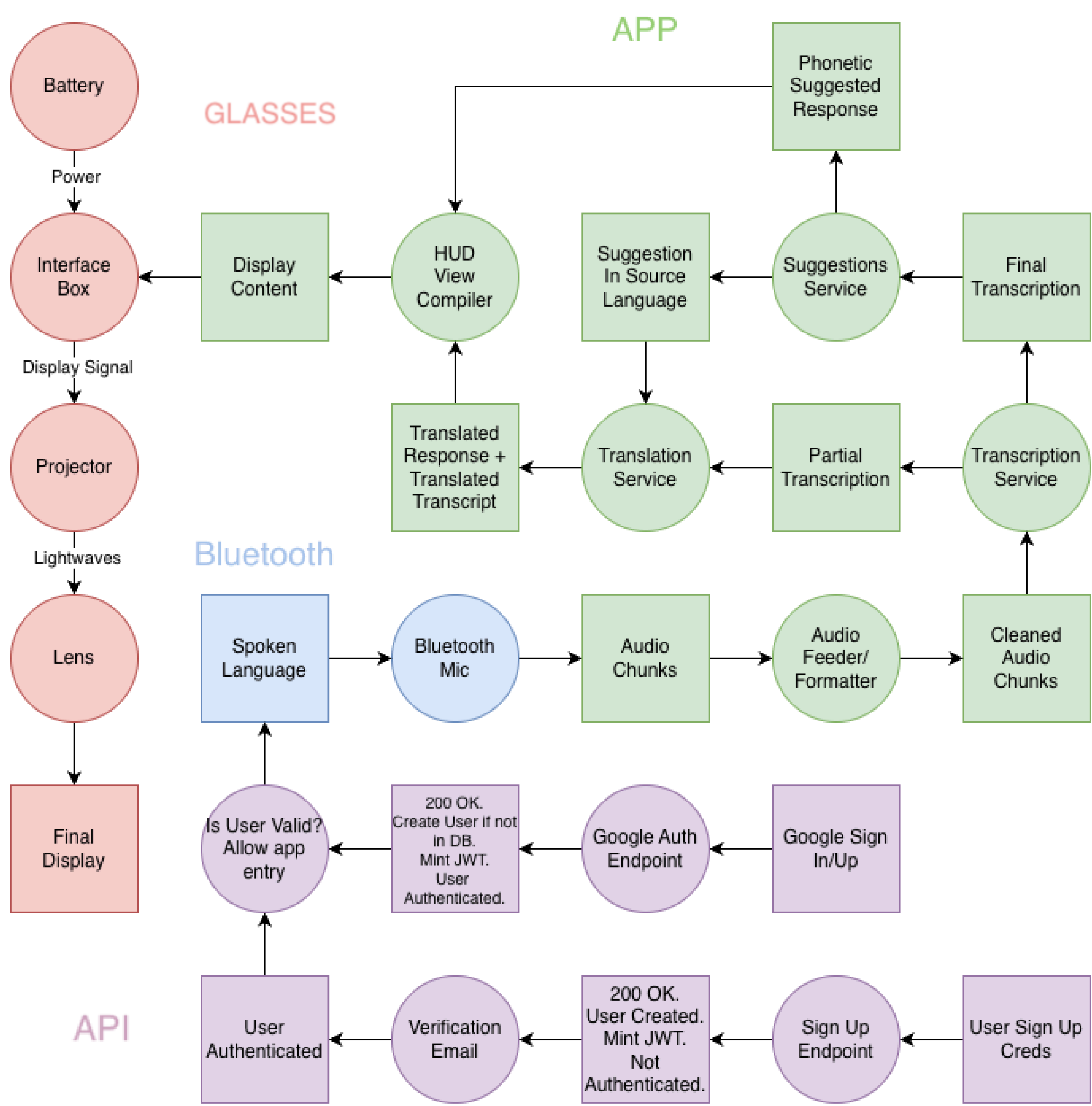


Figure 1: System Block Diagram

Problem Statement

In our increasingly globalizing world, language barriers often limit smooth and natural communication. Existing translation solutions (like phone apps) disrupt the flow of conversation by requiring users to look away, speak into a device, or share a screen. Our design aims to address this issue with glasses that deliver real-time translation directly in the user’s line of view, allowing users to converse seamlessly while maintaining eye contact and natural interaction.

HUDini Glasses

AI-Powered Language Translation

Companion App

- We developed an app that would allow users to monitor their account (accessible using integrated Google Auth), manage their data, and provide user information that helps guide the AI-powered suggested responses.
- Additionally, users can view their previous conversations and use them for language learning (Figure 2).

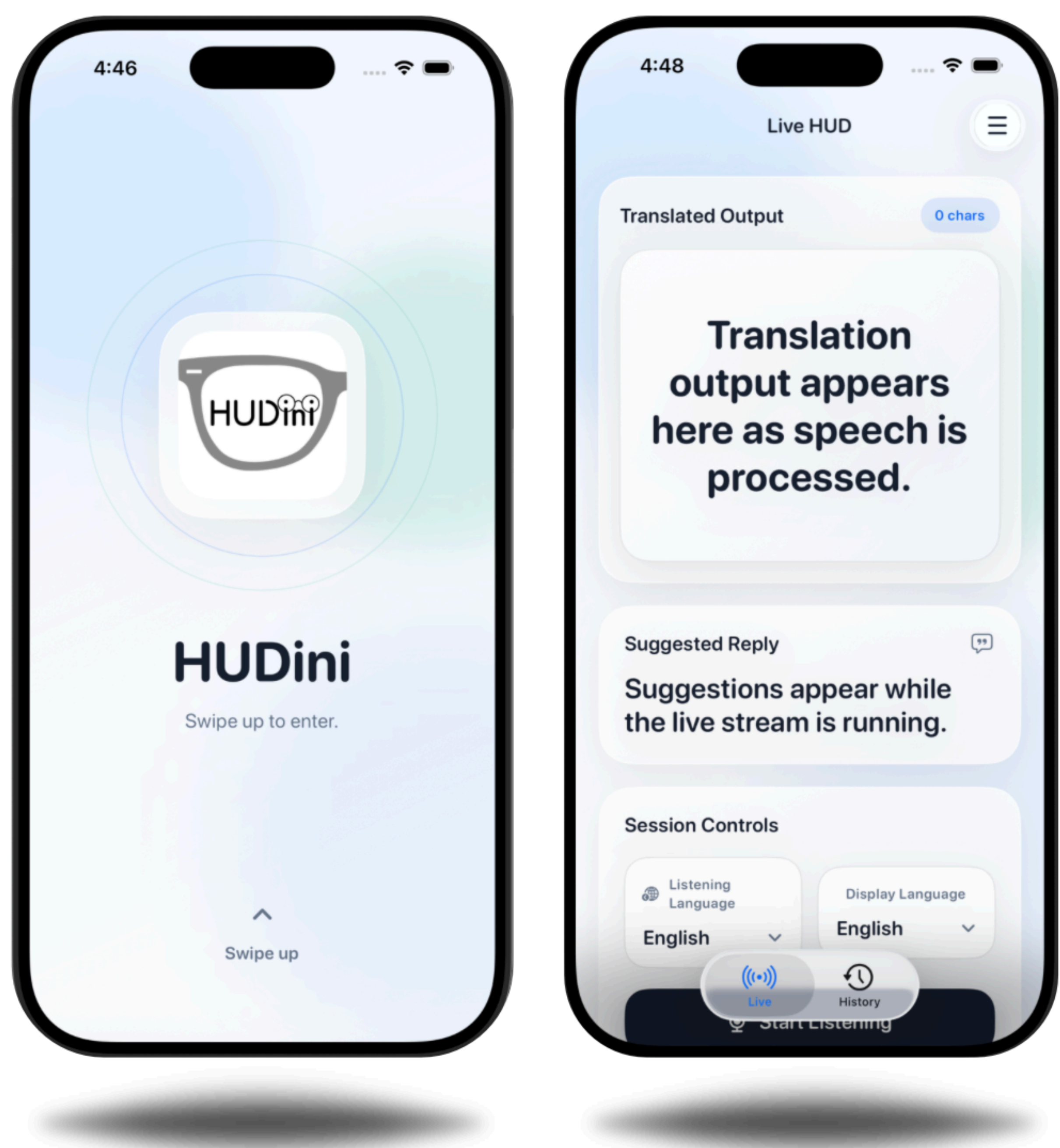


Figure 2: HUDini Companion App

Glasses Frame

- We designed a custom glasses frame to effectively house all components and ensure a light and wearable device.
- We 3D printed the CAD frames to complete a light and low-cost, highly effective device (Figure 3).

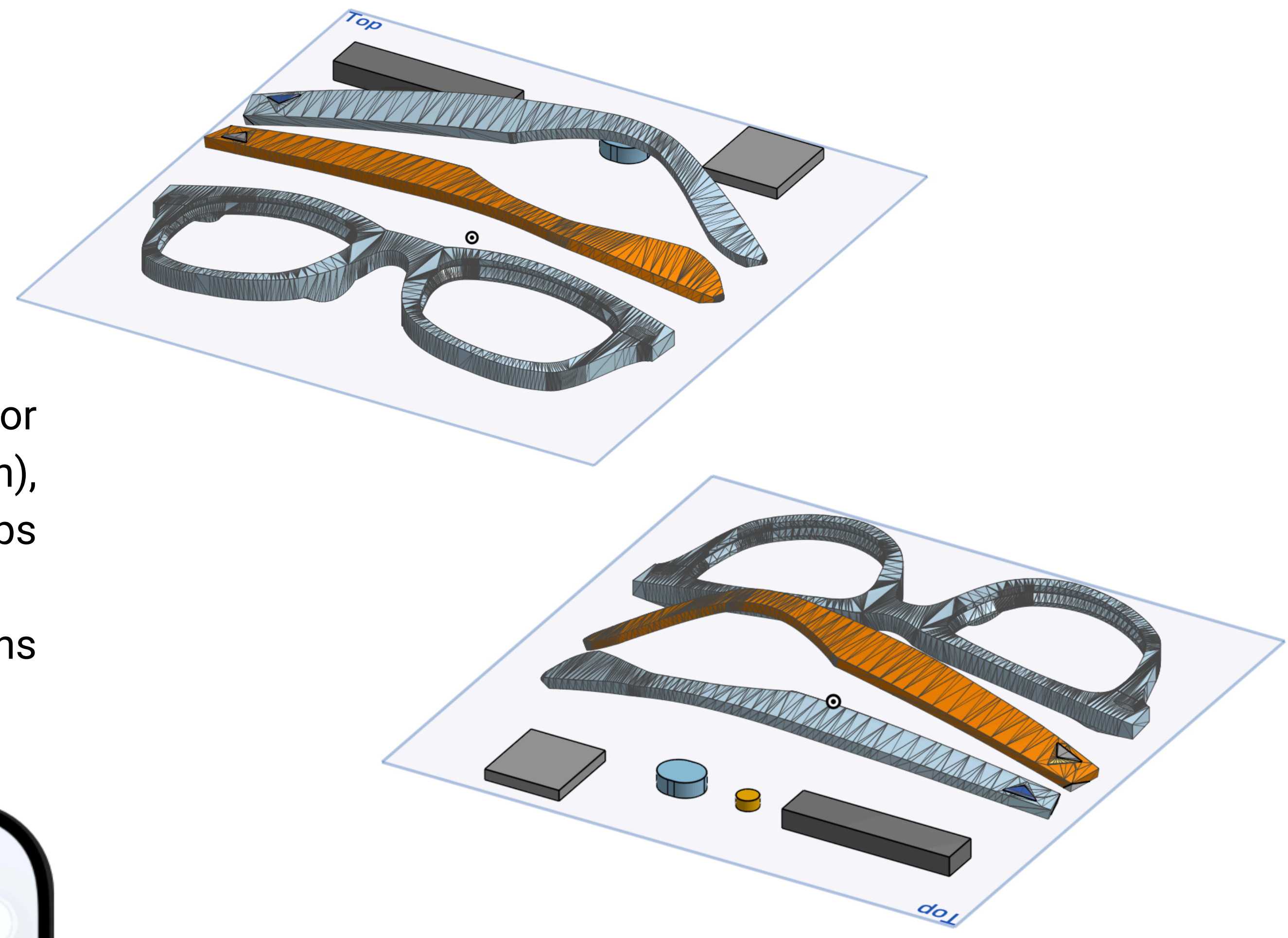


Figure 3: Glasses Housing CAD Design

System Software & Hardware

- Our main challenge was finding a way to limit latency and ensure conversations maintained a natural flow.
- We were able to build a fast and highly responsive system by integrating Apple’s Transcription and Translation tools, as well as OpenAI’s API for live response suggestions.
- Most of the software development was done in Swift, Python, and C++, with the firmware running in Lua.
- When approaching the hardware design of our system, our main aim was to build a minimal system to keep the device light and wearable, and to keep production costs low.
- Thus, we mapped out our functionality priorities and designed our hardware around them, settling on a system that includes a HUD, microphone, battery, and processor.
- We also designed our own custom PCB (Figure 4), which would comfortably fit into our custom glasses frame.

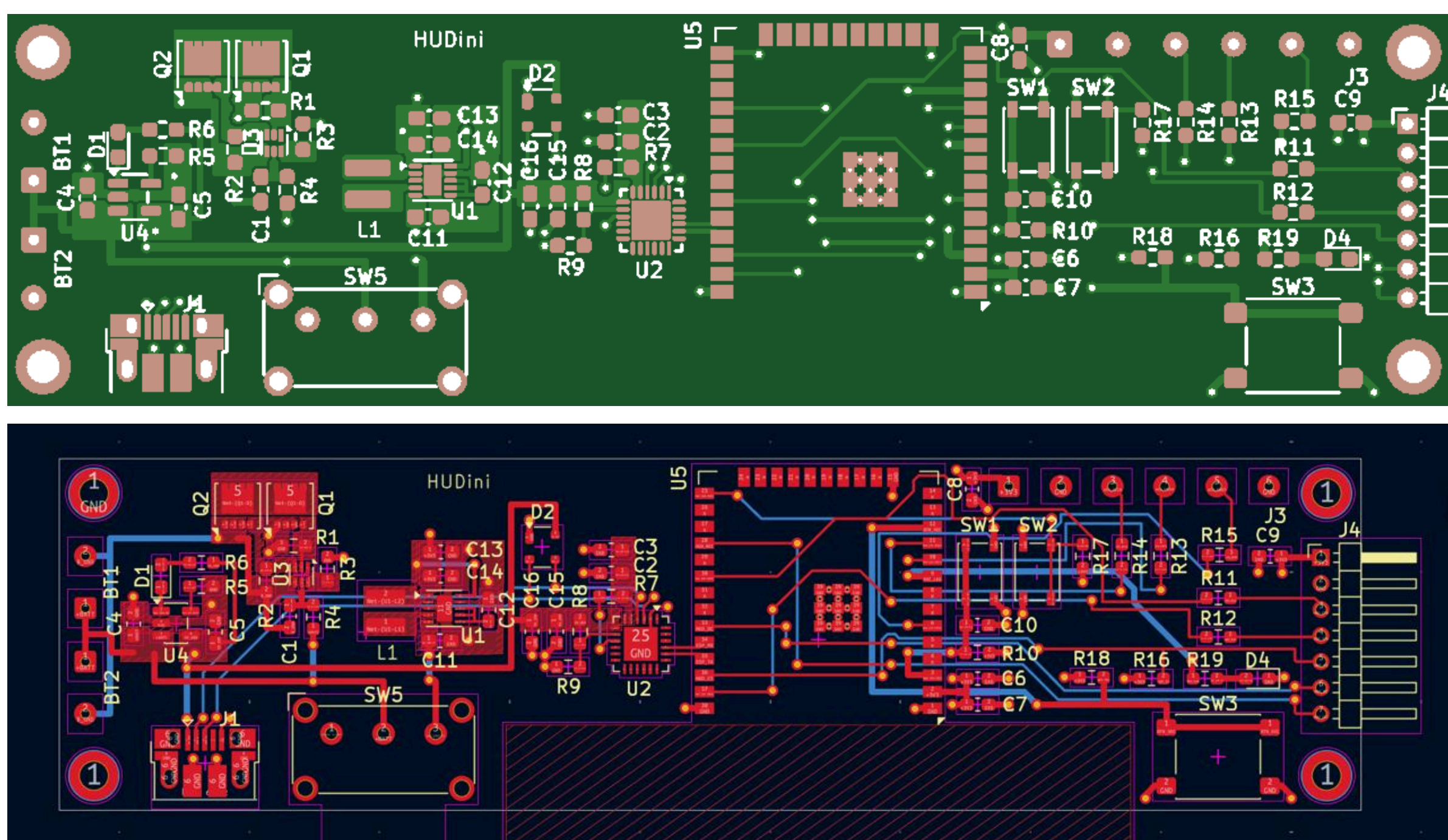


Figure 4: Custom PCB Design

